Worksheet 11 · Quadratic Inequalities IGCSE

Edexcel IGCSE Mathematics · past paper questions

Name:	Subject: Math	Date:
Learning Objectives: Students will be able to: • Solve quadratic inequalities • Use a sketch of the graph to find the solution set • Write solutions using correct inequality notation		
Grade: / 32	**Teacher's Feedback:**	

Answer in the space given. Show your working. Marks for each part are shown in brackets.

Questions *Quadratic Inequalities*

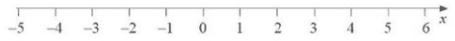
Question 1 (4 marks)

Solve $x^2 + 3x - 10 \leqslant 0$

.....

(3)

(b) Represent your solution set to part (a) on the number line below.



(1)

Question 2 (3 marks)

Solve the inequality $5y^2 - 17y \leqslant 40$

.....

(3)

Question 3 (5 marks)

The diagram shows a rectangle.

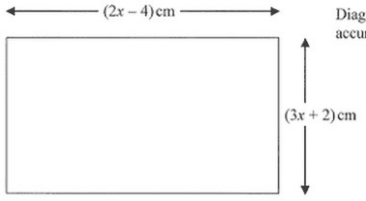


Diagram **NOT** accurately drawn

The area of the rectangle is A cm^2

Given that $A < 3x + 27$
find the range of possible values for x.

.....

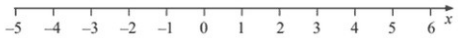
Question 4 (4 marks)

(a) Solve $x^2 + 2x > 6x + 5$

.....

(3)

(b) Represent your solution set to part (a) on the number line below.



(1)

Question 5 (4 marks)

$f(x) = x^2 - 4$

$g(x) = 2x + 1$

Solve $fg(x) > 0$
Show clear algebraic working.

.....

Question 6 (3 marks)

Solve the inequality $4x^2 - 5x - 6 > 0$

Question 7 (4 marks)

n is an integer such that $3n + 2 \leqslant 14$ and $\dfrac{6n}{n^2 + 5} > 1$

Find all the possible values of n.

Question 8 (5 marks)

Here is a rectangle.

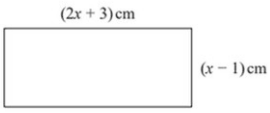


Diagram **NOT** accurately drawn

Given that the area of the rectangle is less than 75 cm²

find the range of possible values of x